

MAT1001 Calculus for Engineers

Triple Integrals, Cylindrical and Spherical Coordinates



VIT-AP
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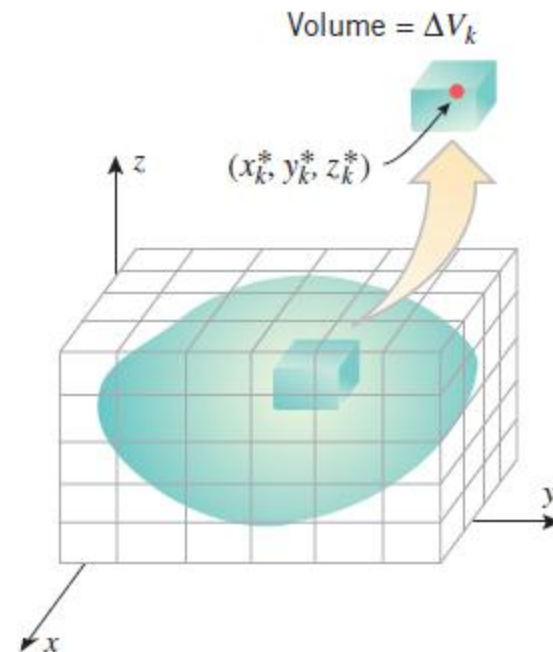
Dr. Aswathy R K

Department of Mathematics

School of Advanced Sciences

Vellore Institute of Technology – Andhra Pradesh

$$\iiint_G f(x, y, z) dV = \lim_{n \rightarrow +\infty} \sum_{k=1}^n f(x_k^*, y_k^*, z_k^*) \Delta V_k$$



Triple Integration - Properties

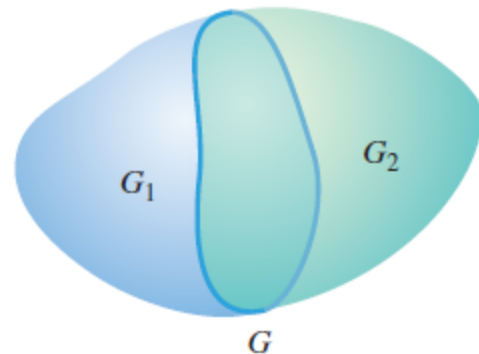
$$\iiint_G cf(x, y, z) dV = c \iiint_G f(x, y, z) dV \quad (c \text{ a constant})$$

$$\iiint_G [f(x, y, z) + g(x, y, z)] dV = \iiint_G f(x, y, z) dV + \iiint_G g(x, y, z) dV$$

$$\iiint_G [f(x, y, z) - g(x, y, z)] dV = \iiint_G f(x, y, z) dV - \iiint_G g(x, y, z) dV$$

Moreover, if the region G is subdivided into two subregions G_1 and G_2 then

$$\iiint_G f(x, y, z) dV = \iiint_{G_1} f(x, y, z) dV + \iiint_{G_2} f(x, y, z) dV$$



Theorem – Integration over a rectangular box

THEOREM (Fubini's Theorem^{*}) *Let G be the rectangular box defined by the inequalities*

$$a \leq x \leq b, \quad c \leq y \leq d, \quad k \leq z \leq l$$

If f is continuous on the region G , then

$$\iiint_G f(x, y, z) dV = \int_a^b \int_c^d \int_k^l f(x, y, z) dz dy dx \quad (2)$$

Moreover, the iterated integral on the right can be replaced with any of the five other iterated integrals that result by altering the order of integration.

Six orders of integration are possible for the iterated integral in Theorem 14.5.1:

$$dx dy dz, \quad dy dz dx, \quad dz dx dy \\ dx dz dy, \quad dz dy dx, \quad dy dx dz$$

Example

- Evaluate the triple integral

$$\iiint_G 12xy^2z^3 dV$$

over the rectangular box G defined by the inequalities $-1 \leq x \leq 2, 0 \leq y \leq 3, 0 \leq z \leq 2$.

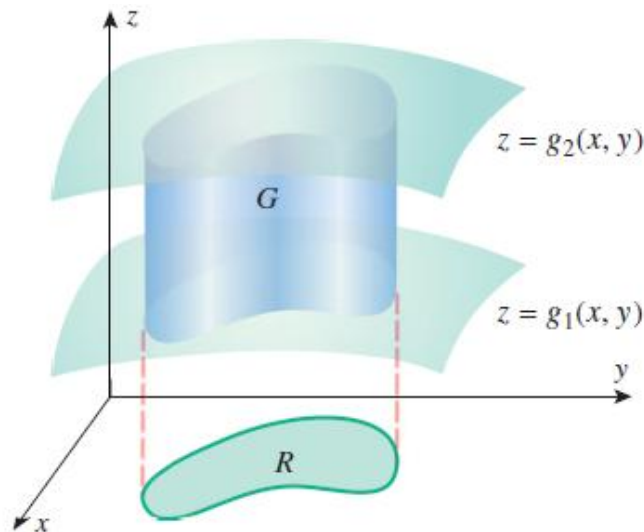
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$$\begin{aligned}\iiint_G 12xy^2z^3 dV &= \int_{-1}^2 \int_0^3 \int_0^2 12xy^2z^3 dz dy dx \\&= \int_{-1}^2 \int_0^3 [3xy^2z^4]_{z=0}^2 dy dx = \int_{-1}^2 \int_0^3 48xy^2 dy dx \\&= \int_{-1}^2 [16xy^3]_{y=0}^3 dx = \int_{-1}^2 432x dx \\&= 216x^2 \Big|_{-1}^2 = 648\end{aligned}$$

Theorem - Integration over more general regions

THEOREM Let G be a simple xy -solid with upper surface $z = g_2(x, y)$ and lower surface $z = g_1(x, y)$, and let R be the projection of G on the xy -plane. If $f(x, y, z)$ is continuous on G , then

$$\iiint_G f(x, y, z) dV = \iint_R \left[\int_{g_1(x, y)}^{g_2(x, y)} f(x, y, z) dz \right] dA \quad (3)$$



$g_1(x, y)$ and $g_2(x, y)$ are continuous on R

$g_1(x, y) \leq g_2(x, y)$ on R .

R is type I or type II region

} *simple xy -solid.*

Determining Limits of Integration: Simple xy -Solid

- Step 1.** Find an equation $z = g_2(x, y)$ for the upper surface and an equation $z = g_1(x, y)$ for the lower surface of G . The functions $g_1(x, y)$ and $g_2(x, y)$ determine the lower and upper z -limits of integration.
- Step 2.** Make a two-dimensional sketch of the projection R of the solid on the xy -plane. From this sketch determine the limits of integration for the double integral over R in (3).

Example

- Let G be the wedge in the first octant that is cut from the cylindrical solid $y^2 + z^2 \leq 1$ by the planes $y = x$ and $x = 0$. Evaluate

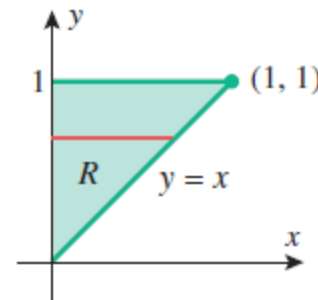
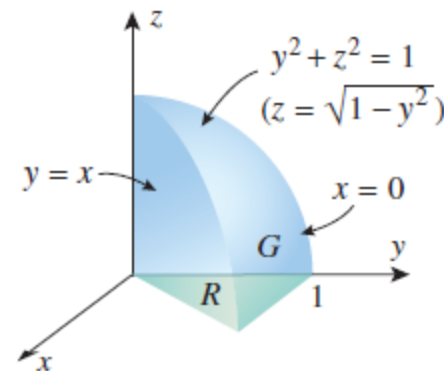
$$\iiint_G z \, dV$$

$$\Rightarrow \iiint_G z \, dV = \iint_R \left[\int_0^{\sqrt{1-y^2}} z \, dz \right] dA$$

$$\iiint_G z \, dV = \int_0^1 \int_0^y \int_0^{\sqrt{1-y^2}} z \, dz \, dx \, dy = \int_0^1 \int_0^y \left[\frac{1}{2} z^2 \right]_{z=0}^{\sqrt{1-y^2}} dx \, dy$$

$$= \int_0^1 \int_0^y \frac{1}{2} (1 - y^2) \, dx \, dy = \frac{1}{2} \int_0^1 (1 - y^2) x \Big|_{x=0}^y dy$$

$$= \frac{1}{2} \int_0^1 (y - y^3) \, dy = \frac{1}{2} \left[\frac{1}{2} y^2 - \frac{1}{4} y^4 \right]_0^1 = \frac{1}{8}$$

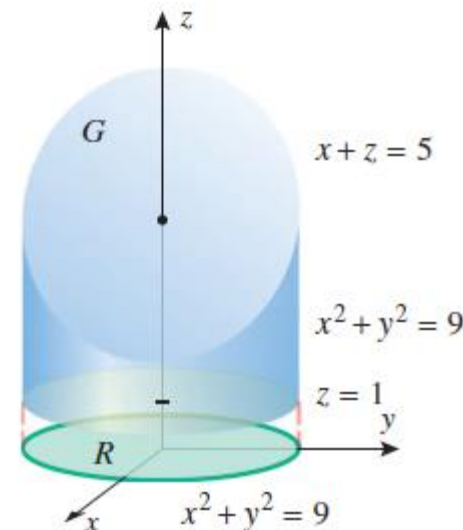
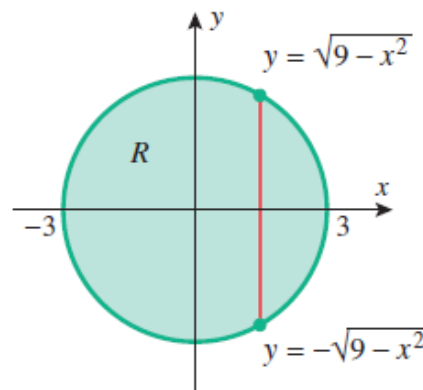


Volume as triple integrals

$$\text{volume of } G = \iiint_G dV$$

- Use a triple integral to find the volume of the solid within the cylinder $x^2 + y^2 = 9$ and between the planes $z = 1$ and $x + z = 5$.

$$\begin{aligned} \Rightarrow \text{volume of } G &= \iiint_G dV = \iint_R \left[\int_1^{5-x} dz \right] dA \\ &= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_1^{5-x} dz \, dy \, dx \\ &= \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} (4-x) \, dy \, dx = 36\pi \end{aligned}$$



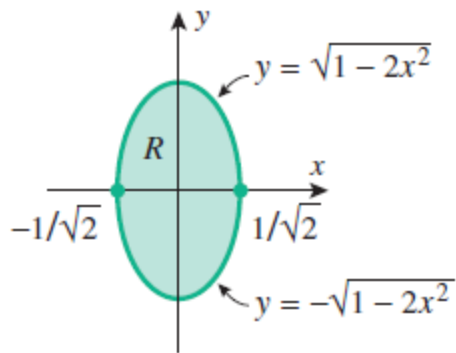
Example

- Find the volume of the solid enclosed between the paraboloids

$$z = 5x^2 + 5y^2 \quad \text{and} \quad z = 6 - 7x^2 - y^2$$

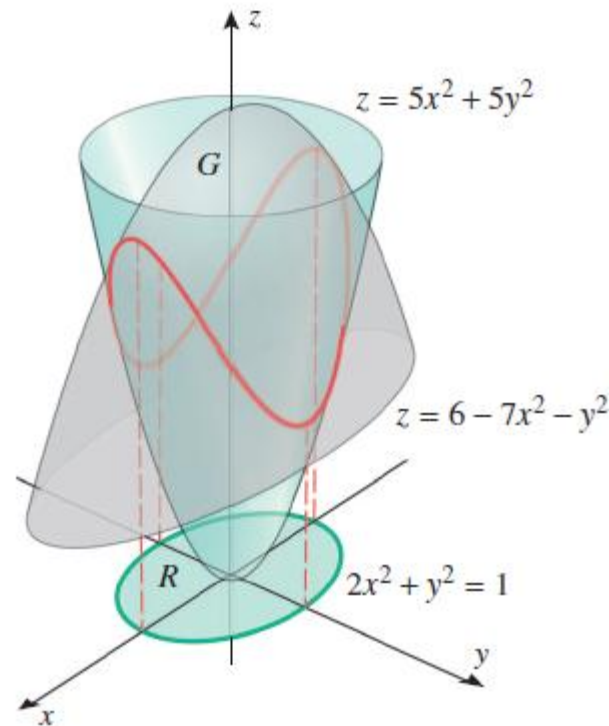
$$\rightarrow \text{volume of } G = \iiint_G dV = \iint_R \left[\int_{5x^2+5y^2}^{6-7x^2-y^2} dz \right] dA$$

The projection R is obtained by solving the given equations



$$5x^2 + 5y^2 = 6 - 7x^2 - y^2$$

$$2x^2 + y^2 = 1$$

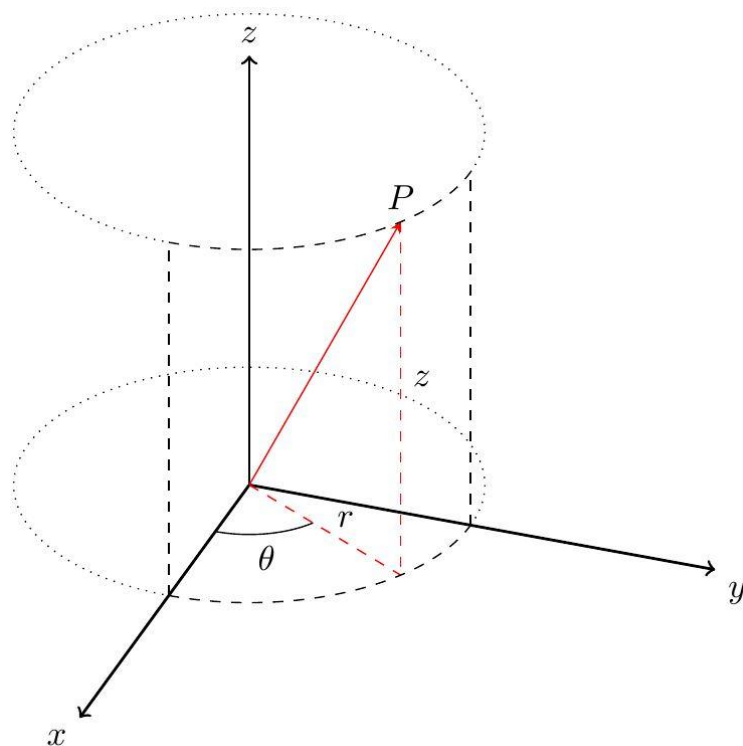


Example

- Find the volume of the solid enclosed between the paraboloids

$$z = 5x^2 + 5y^2 \quad \text{and} \quad z = 6 - 7x^2 - y^2$$

$$\begin{aligned} \rightarrow \text{volume of } G &= \iiint_G dV = \iint_R \left[\int_{5x^2+5y^2}^{6-7x^2-y^2} dz \right] dA \\ &= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} \int_{5x^2+5y^2}^{6-7x^2-y^2} dz \, dy \, dx \\ &= \int_{-1/\sqrt{2}}^{1/\sqrt{2}} \int_{-\sqrt{1-2x^2}}^{\sqrt{1-2x^2}} (6 - 12x^2 - 6y^2) \, dy \, dx \\ &= \frac{3\pi}{\sqrt{2}} \end{aligned}$$



$$x = r \cos \theta$$

$$y = r \sin \theta$$

$$z = z$$

$$r = \sqrt{x^2 + y^2}$$

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

$$z = z$$

Triple integration and Cylindrical coordinates

THEOREM Let G be a solid region whose upper surface has the equation $z = g_2(r, \theta)$ and whose lower surface has the equation $z = g_1(r, \theta)$ in cylindrical coordinates. If the projection of the solid on the xy -plane is a simple polar region R , and if $f(r, \theta, z)$ is continuous on G , then

$$\iiint_G f(r, \theta, z) dV = \iint_R \left[\int_{g_1(r, \theta)}^{g_2(r, \theta)} f(r, \theta, z) dz \right] dA \quad (4)$$

where the double integral over R is evaluated in polar coordinates. In particular, if the projection R is as shown in Figure 14.6.4, then (4) can be written as

$$\iiint_G f(r, \theta, z) dV = \int_{\theta_1}^{\theta_2} \int_{r_1(\theta)}^{r_2(\theta)} \int_{g_1(r, \theta)}^{g_2(r, \theta)} f(r, \theta, z) r dz dr d\theta \quad (5)$$

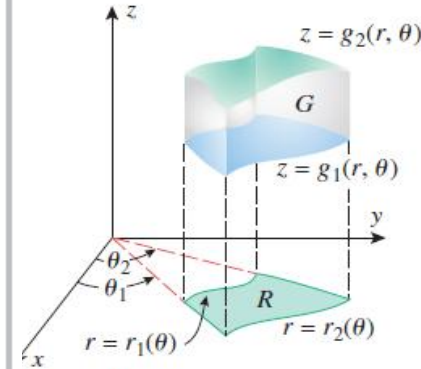


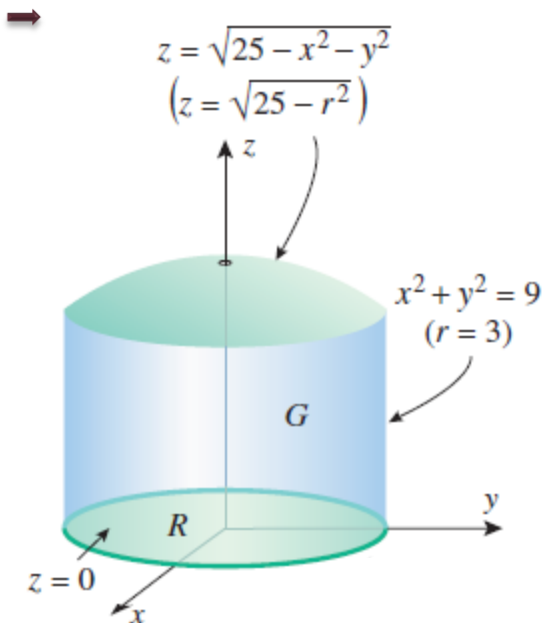
Figure 14.6.4

Determining Limits of Integration: Cylindrical Coordinates

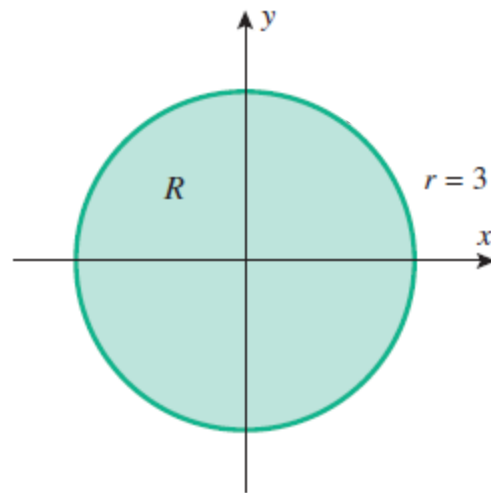
- Step 1.** Identify the upper surface $z = g_2(r, \theta)$ and the lower surface $z = g_1(r, \theta)$ of the solid. The functions $g_1(r, \theta)$ and $g_2(r, \theta)$ determine the z -limits of integration. (If the upper and lower surfaces are given in rectangular coordinates, convert them to cylindrical coordinates.)
- Step 2.** Make a two-dimensional sketch of the projection R of the solid on the xy -plane. From this sketch the r - and θ -limits of integration may be obtained exactly as with double integrals in polar coordinates.

Example

- Use triple integration in cylindrical coordinates to find the volume of the solid G that is bounded above by the hemisphere $z = \sqrt{25 - x^2 - y^2}$, below by the xy -plane, and laterally by the cylinder $x^2 + y^2 = 9$.



$$\begin{aligned}
 V &= \iiint_G dV = \iint_R \left[\int_0^{\sqrt{25-r^2}} dz \right] dA \\
 &= \int_0^{2\pi} \int_0^3 \int_0^{\sqrt{25-r^2}} r \, dz \, dr \, d\theta \\
 &= \int_0^{2\pi} \int_0^3 r \sqrt{25-r^2} \, dr \, d\theta \\
 &= \int_0^{2\pi} \frac{61}{3} \, d\theta = \frac{122}{3} \pi
 \end{aligned}$$



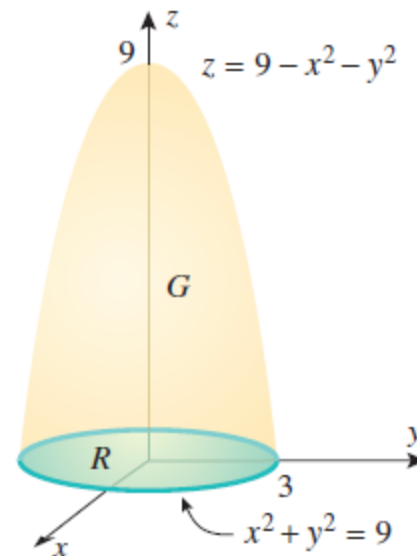
$$\iiint_G f(x, y, z) dV = \iiint_{\text{appropriate limits}} f(r \cos \theta, r \sin \theta, z) r dz dr d\theta$$

- Use cylindrical coordinates to evaluate

$$\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} x^2 dz dy dx$$

→

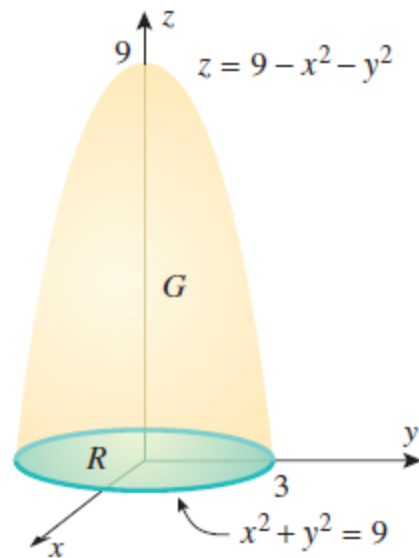
$$\begin{aligned} \int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} x^2 dz dy dx &= \iiint_G x^2 dV \\ &= \iint_R \left[\int_0^{9-r^2} r^2 \cos^2 \theta dz \right] dA = \int_0^{2\pi} \int_0^3 \int_0^{9-r^2} (r^2 \cos^2 \theta) r dz dr d\theta \end{aligned}$$



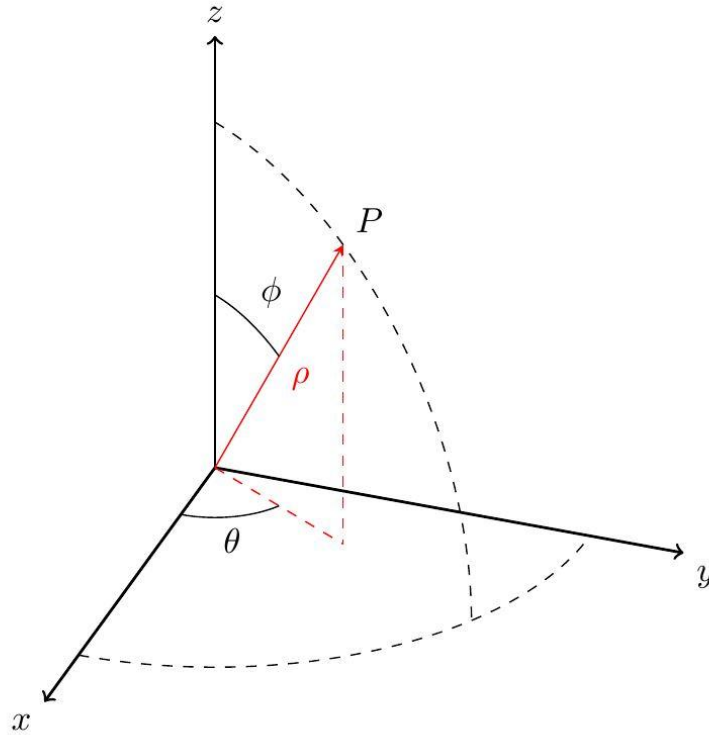
Conversion of triple integration to cylindrical coordinates



$$\begin{aligned}\int_{-3}^3 \int_{-\sqrt{9-x^2}}^{\sqrt{9-x^2}} \int_0^{9-x^2-y^2} x^2 dz dy dx &= \iiint_G x^2 dV \\&= \iiint_R \left[\int_0^{9-r^2} r^2 \cos^2 \theta dz \right] dA = \int_0^{2\pi} \int_0^3 \int_0^{9-r^2} (r^2 \cos^2 \theta) r dz dr d\theta \\&= \int_0^{2\pi} \int_0^3 \int_0^{9-r^2} r^3 \cos^2 \theta dz dr d\theta = \int_0^{2\pi} \int_0^3 [zr^3 \cos^2 \theta]_{z=0}^{9-r^2} dr d\theta \\&= \int_0^{2\pi} \int_0^3 (9r^3 - r^5) \cos^2 \theta dr d\theta = \int_0^{2\pi} \left[\left(\frac{9r^4}{4} - \frac{r^6}{6} \right) \cos^2 \theta \right]_{r=0}^3 d\theta \\&= \frac{243}{4} \int_0^{2\pi} \cos^2 \theta d\theta = \frac{243}{4} \int_0^{2\pi} \frac{1}{2} (1 + \cos 2\theta) d\theta = \frac{243\pi}{4}\end{aligned}$$



Spherical coordinates and Triple integration



$$0 \leq \theta \leq 2\pi, \quad 0 \leq \phi \leq \pi, \quad \rho \geq 0$$

$$x = \rho \cos \theta \sin \phi$$

$$y = \rho \sin \theta \sin \phi$$

$$z = \rho \cos \phi$$

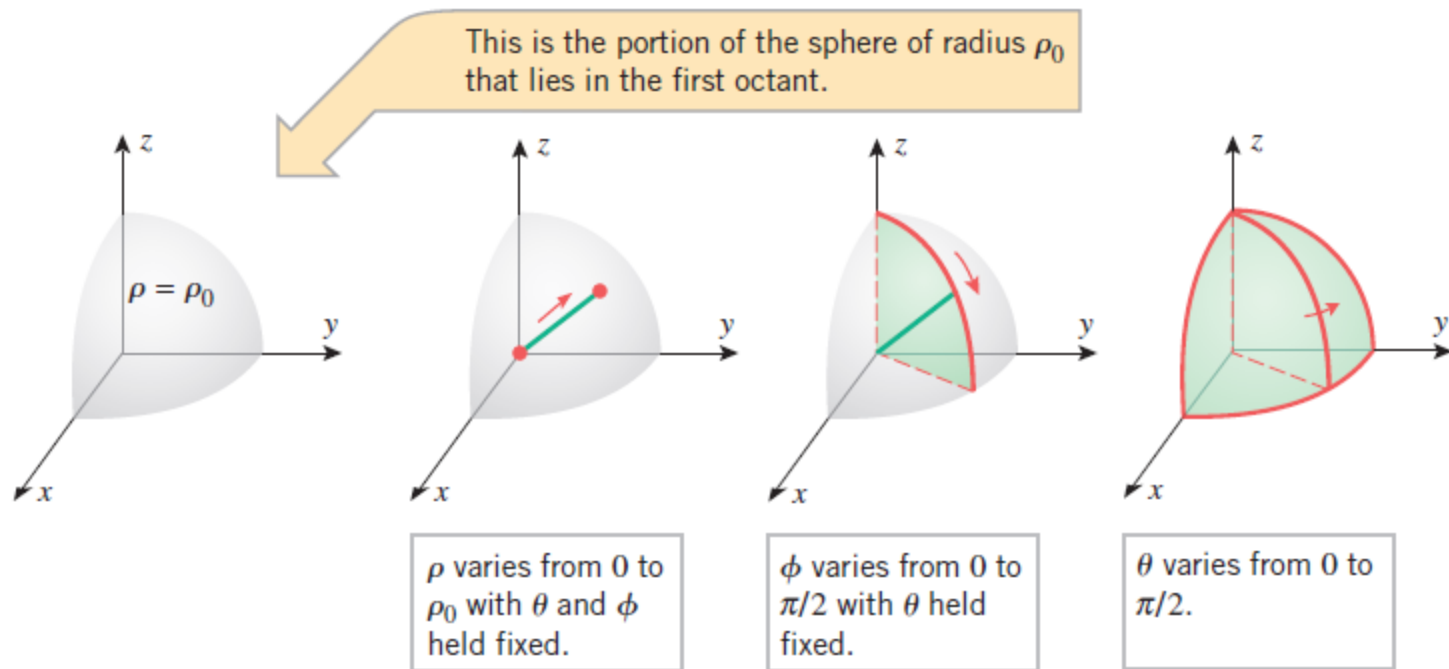
$$\rho = \sqrt{x^2 + y^2 + z^2}$$

$$\theta = \tan^{-1} \left(\frac{y}{x} \right)$$

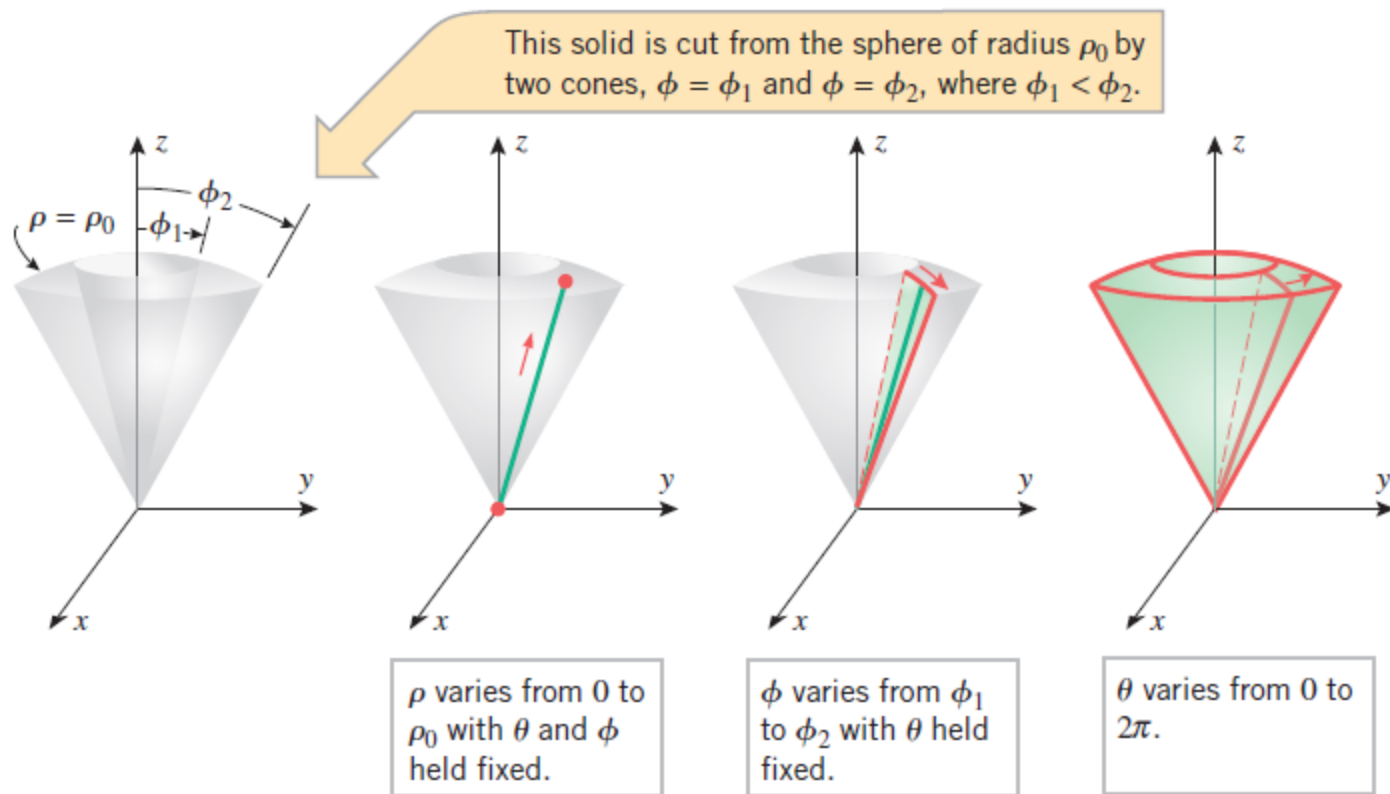
$$\phi = \cos^{-1} \left(\frac{z}{\rho} \right)$$

$$\iiint_G f(\rho, \theta, \phi) dV = \iiint_{\text{appropriate limits}} f(\rho, \theta, \phi) \rho^2 \sin \phi d\rho d\phi d\theta$$

Examples of finding limits

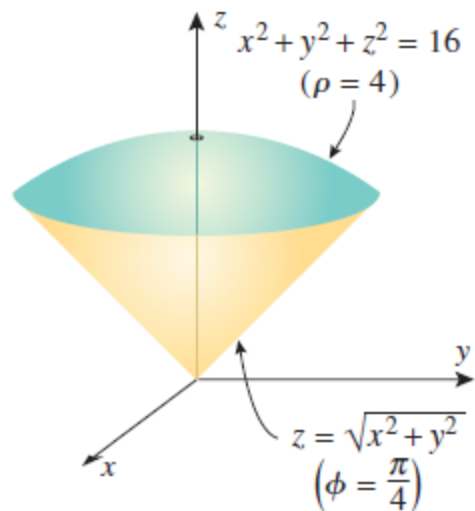


Examples of finding limits



Example

- Use spherical coordinates to find the volume of the solid G bounded above by the sphere $x^2 + y^2 + z^2 = 16$ and below by the cone $z = \sqrt{x^2 + y^2}$.



$$z = \sqrt{x^2 + y^2}$$

$$\rho \cos \phi = \sqrt{\rho^2 \sin^2 \phi \cos^2 \theta + \rho^2 \sin^2 \phi \sin^2 \theta}$$

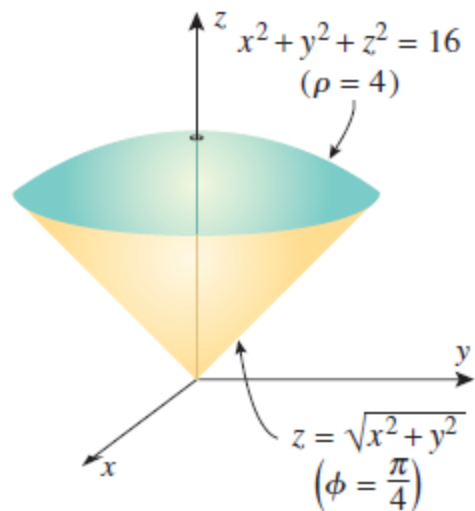
$$\rho \cos \phi = \rho \sin \phi$$

$$\tan \phi = 1,$$

$$\phi = \pi/4$$

Example

- Use spherical coordinates to find the volume of the solid G bounded above by the sphere $x^2 + y^2 + z^2 = 16$ and below by the cone $z = \sqrt{x^2 + y^2}$.



$$\begin{aligned}
 V &= \iiint_G dV = \int_0^{2\pi} \int_0^{\pi/4} \int_0^4 \rho^2 \sin \phi \, d\rho \, d\phi \, d\theta \\
 &= \int_0^{2\pi} \int_0^{\pi/4} \left[\frac{\rho^3}{3} \sin \phi \right]_{\rho=0}^4 d\phi \, d\theta \\
 &= \int_0^{2\pi} \int_0^{\pi/4} \frac{64}{3} \sin \phi \, d\phi \, d\theta \\
 &= \frac{64}{3} \int_0^{2\pi} [-\cos \phi]_{\phi=0}^{\pi/4} d\theta = \frac{64}{3} \int_0^{2\pi} \left(1 - \frac{\sqrt{2}}{2} \right) d\theta \\
 &= \frac{64\pi}{3} (2 - \sqrt{2})
 \end{aligned}$$

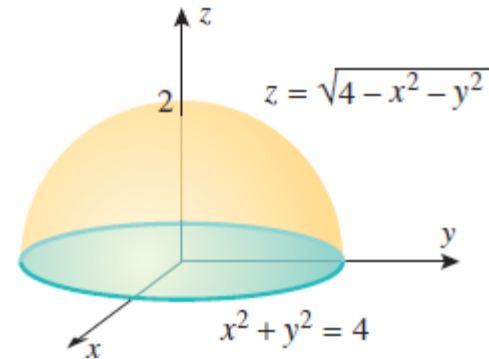
$$\iiint_G f(x, y, z) dV = \iiint_{\text{appropriate limits}} f(\rho \sin \phi \cos \theta, \rho \sin \phi \sin \theta, \rho \cos \phi) \rho^2 \sin \phi d\rho d\phi d\theta$$

- Use spherical coordinates to evaluate

$$\int_{-2}^2 \int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}} \int_0^{\sqrt{4-x^2-y^2}} z^2 \sqrt{x^2 + y^2 + z^2} dz dy dx$$

→

$$\begin{aligned} \iiint_G z^2 \sqrt{x^2 + y^2 + z^2} dV &= \int_0^{2\pi} \int_0^{\pi/2} \int_0^2 \rho^5 \cos^2 \phi \sin \phi d\rho d\phi d\theta \\ &= \int_0^{2\pi} \int_0^{\pi/2} \frac{32}{3} \cos^2 \phi \sin \phi d\phi d\theta \\ &= \frac{32}{3} \int_0^{2\pi} \left[-\frac{1}{3} \cos^3 \phi \right]_{\phi=0}^{\pi/2} d\theta = \frac{32}{9} \int_0^{2\pi} d\theta = \frac{64}{9} \pi \end{aligned}$$



Thank You...